1.

```
MariaDB [employee_dump]> select e1.name, e2.name as manager from employees as e1 join employees as e2 on e1.manager_id = e2.id;
+-----------+----------+
| name      | manager  |
+-----------+----------+
| Charlotte | Emma     |
| Ella      | William  |
| Scarlett  | Olivia   |
| Ava       | Ethan    |
| Scarlett  | Sophia   |
| Charlotte | Robert   |
| Evelyn    | Ava      |
| Mia       | Noah     |
| Ava       | Ethan    |
| Ella      | James    |
| Logan     | Emma     |
| Evelyn    | Emma     |
| Scarlett  | Ethan    |
| Zoe       | Noah     |
| Mia       | Isabella |
| Grace     | Ethan    |
| Olivia    | James    |
| Scarlett  | William  |
| Mia       | Ava      |
| Emma      | Olivia   |
| Ethan     | Ethan    |
| Logan     | Maria    |
| Abigail   | Emma     |
| Aiden     | Noah     |
| Ava       | Liam     |
| Charlotte | James    |
| Mateo     | Sophia   |
| Charlotte | Olivia   |
| Jackson   | Sophia   |
| Evelyn    | Ethan    |
| Joseph    | James    |
| Amelia    | Liam     |
| Elijah    | William  |
| Grace     | Liam     |
| Olivia    | Isabella |
| Aiden     | Emma     |
| Mason     | Maria    |
| Ava       | Ethan    |
| Ava       | Liam     |
| Ava       | Emma     |
| Elijah    | Emma     |
| Sophia    | Maria    |
| Joseph    | Ava      |
| Oliver    | Ava      |
| Mia       | Liam     |
| Zoe       | Maria    |
| Jackson   | James    |
| Amelia    | Noah     |
| Ella      | Emma     |
| Evelyn    | Olivia   |
| Grace     | William  |
| Isabella  | Robert   |
| Scarlett  | Noah     |
| Joseph    | James    |
```

2.

```
MariaDB [employee_dump]> select departments.department_name, employees.name as manager from d
epartments left join employees on departments.manager_id = employees.manager_id;
+------------------+-----------+
| department_name  | manager   |
+------------------+-----------+
| Assembly         | Ella      |
| Assembly         | Olivia    |
| Assembly         | Charlotte |
| Assembly         | Joseph    |
| Assembly         | Jackson   |
| Assembly         | Joseph    |
| Assembly         | Aiden     |
| Assembly         | Amelia    |
| Assembly         | Ava       |
| Assembly         | Sophia    |
| Assembly         | Ethan     |
| Assembly         | Zoe       |
| Assembly         | Logan     |
| Assembly         | Charlotte |
| Assembly         | David     |
| Assembly         | Logan     |
| Assembly         | Charlotte |
| Assembly         | Ella      |
| Assembly         | Lucas     |
| Quality Control  | Logan     |
| Quality Control  | Mason     |
| Quality Control  | Sophia    |
| Quality Control  | Zoe       |
| Quality Control  | Mason     |
| Quality Control  | Sophia    |
| Quality Control  | Oliver    |
| Quality Control  | Isabella  |
| Machining        | Charlotte |
| Machining        | Isabella  |
| Machining        | Mia       |
| Machining        | Oliver    |
| Machining        | Lucas     |
| Machining        | Emma      |
| Machining        | Emma      |
| Machining        | Ethan     |
| Machining        | Mia       |
| Machining        | Grace     |
| Logistics        | Charlotte |
| Logistics        | Logan     |
| Logistics        | Evelyn    |
| Logistics        | Abigail   |
| Logistics        | Aiden     |
| Logistics        | Ava       |
| Logistics        | Elijah    |
| Logistics        | Ella      |
| Logistics        | Grace     |
| Logistics        | Elijah    |
| Logistics        | Mia       |
| Logistics        | Olivia    |
| Logistics        | Liam      |
| Logistics        | Emma      |
| Maintenance      | Ava       |
```

3.

```
MariaDB [employee_dump]>  select name, departments.department_name from employees left join d
epartments on employees.department_id = departments.id;
+-----------+-----------------+
| name      | department_name |
+-----------+-----------------+
| James     | Assembly        |
| Maria     | Quality Control |
| Robert    | Machining       |
| Emma      | Logistics       |
| Liam      | Maintenance     |
| Olivia    | R&D             |
| Noah      | Procurement     |
| Ava       | Health & Safety |
| William   | Packaging       |
| Sophia    | Inventory       |
| Ethan     | Welding         |
| Isabella  | Plant Management |
| Charlotte | Machining       |
| Ella      | Health & Safety |
| Scarlett  | Health & Safety |
| Ava       | Procurement     |
| Scarlett  | R&D             |
| Charlotte | Inventory       |
| Evelyn    | Machining       |
| Mia       | NULL            |
| Ava       | R&D             |
| Ella      | Procurement     |
| Logan     | Health & Safety |
| Evelyn    | R&D             |
| Scarlett  | Assembly        |
| Zoe       | R&D             |
| Mia       | Plant Management |
| Grace     | Procurement     |
| Olivia    | Assembly        |
| Scarlett  | Quality Control |
| Mia       | Plant Management |
| Emma      | Assembly        |
| Ethan     | Assembly        |
| Logan     | Health & Safety |
| Abigail   | Plant Management |
| Aiden     | Welding         |
| Ava       | NULL            |
| Charlotte | Quality Control |
| Mateo     | Machining       |
| Charlotte | Machining       |
| Jackson   | Packaging       |
| Evelyn    | Quality Control |
| Joseph    | Welding         |
| Amelia    | Plant Management |
| Elijah    | Welding         |
| Grace     | Plant Management |
| Olivia    | Packaging       |
| Aiden     | Logistics       |
| Mason     | Quality Control |
| Ava       | Quality Control |
| Ava       | Welding         |
| Ava       | Inventory       |
```

4.

```
MariaDB [employee_dump]> select name from employees where manager_id is null;
+----------+
| name     |
+----------+
| James    |
| Maria    |
| Robert   |
| Emma     |
| Liam     |
| Olivia   |
| Noah     |
| Ava      |
| William  |
| Sophia   |
| Ethan    |
| Isabella |
| Omar     |
| Sherif   |
+----------+
14 rows in set (0.006 sec)
```

5.

```
MariaDB [employee_dump]> select name from employees where department_id is null;
+-----------+
| name      |
+-----------+
| Mia       |
| Ava       |
| Oliver    |
| Omar      |
| Aiden     |
| Olivia    |
| Logan     |
| Mateo     |
| Sherif    |
| Sebastian |
| Ava       |
| Charlotte |
+-----------+
12 rows in set (0.001 sec)
```

6.

```
MariaDB [employee_dump]> select department_id, count(*) from employees left join departments
on employees.department_id = departments.department_name group by department_id order by depa
rtment_id;
+---------------+----------+
| department_id | count(*) |
+---------------+----------+
|          NULL |       12 |
|             1 |       16 |
|             2 |       12 |
|             3 |       13 |
|             4 |       13 |
|             5 |       10 |
|             6 |       15 |
|             7 |       13 |
|             8 |       24 |
|             9 |       11 |
|            10 |       16 |
|            11 |       17 |
|            12 |       16 |
+---------------+----------+
```

7.

```
MariaDB [employee_dump]> select departments.department_name, employees.name, employees.salary from departments left join employees on departments.manager_id = employees.manager_id;
+-----------------+----------+--------+
| department_name | name     | salary |
+-----------------+----------+--------+
| Assembly        | Ella     | 97000  |
| Assembly        | Olivia   | 50217  |
| Assembly        | Charlotte | 93743 |
| Assembly        | Joseph   | 40289  |
| Assembly        | Jackson  | 54317  |
| Assembly        | Joseph   | 45471  |
| Assembly        | Aiden    | 69245  |
| Assembly        | Amelia   | 63194  |
| Assembly        | Ava      | 51962  |
| Assembly        | Sophia   | 66165  |
| Assembly        | Ethan    | 45847  |
| Assembly        | Zoe      | 69673  |
| Assembly        | Logan    | 48130  |
| Assembly        | Charlotte | 46826 |
| Assembly        | David    | 57431  |
| Assembly        | Logan    | 68598  |
| Assembly        | Charlotte | 44077 |
| Assembly        | Ella     | 52052  |
| Assembly        | Lucas    | 57769  |
| Quality Control | Logan    | 48475  |
| Quality Control | Mason    | 64564  |
| Quality Control | Sophia   | 50018  |
| Quality Control | Zoe      | 44635  |
| Quality Control | Mason    | 51451  |
| Quality Control | Sophia   | 65148  |
| Quality Control | Oliver   | 56634  |
| Quality Control | Isabella | 58331  |
| Machining       | Charlotte | 44715 |
| Machining       | Isabella | 41217  |
| Machining       | Mia      | 61365  |
| Machining       | Oliver   | 42345  |
| Machining       | Lucas    | 55133  |
| Machining       | Emma     | 50688  |
| Machining       | Emma     | 58980  |
| Machining       | Ethan    | 43583  |
| Machining       | Mia      | 45715  |
| Machining       | Grace    | 63678  |
| Logistics       | Charlotte | 53936 |
| Logistics       | Logan    | 69115  |
| Logistics       | Evelyn   | 80367  |
| Logistics       | Abigail  | 50926  |
| Logistics       | Aiden    | 69333  |
| Logistics       | Ava      | 65116  |
| Logistics       | Elijah   | 42566  |
| Logistics       | Ella     | 58686  |
| Logistics       | Grace    | 61983  |
| Logistics       | Elijah   | 50289  |
| Logistics       | Mia      | 58989  |
| Logistics       | Olivia   | 60497  |
| Logistics       | Liam     | 40963  |
| Logistics       | Emma     | 57114  |
| Maintenance     | Ava      | 50089  |
| Maintenance     | Amelia   | 68617  |
```

8.

```
MariaDB [employee_dump]> select e.id, e.name, e.salary, m.name as manager_name, m.salary as m
anager_salary from employees e join employees m on e.manager_id = m.id where e.salary > m.sal
ary;
+----+----------+--------+--------------+----------------+
| id | name     | salary | manager_name | manager_salary |
+----+----------+--------+--------------+----------------+
| 22 | Ella     |  97000 | James        |          95000 |
| 31 | Scarlett |  77000 | William      |          75000 |
+----+----------+--------+--------------+----------------+
2 rows in set (0.007 sec)
```